Microplastic Pollution in Freshwater

Four-Subtopic Literature Map and Method–Effect Evidence Tables

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Abstract

This draft summarises four overlapping sets derived from 10,833 preliminary Web of Science–Scopus include candidates: Causes (6,430), Water Effects (7,575), Laboratory Methods (6,654) and Data Trends (5,320). Tables and PGFPlots charts list every *defined* subcategory and its automated title/abstract match frequency. A second table under each theme gives examples of methods and abstract-reported findings with indexed IEEE citations. The review supports planning a reference-laboratory validation strategy; it is **not a final full-text systematic review**. Subcategory frequencies are multi-label and must not be interpreted as experimentally established effects.

1 Literature identification and classification

From 18,942 database records (8,923 Web of Science; 10,019 Scopus), deduplication removed 7,318 records and document-type filtering excluded 727. The remaining 10,897 records yielded 10,833 preliminary include candidates and 64 requiring manual review. Four overlapping top-level themes were assigned to candidate papers; these are not four mutually exclusive bodies of evidence.

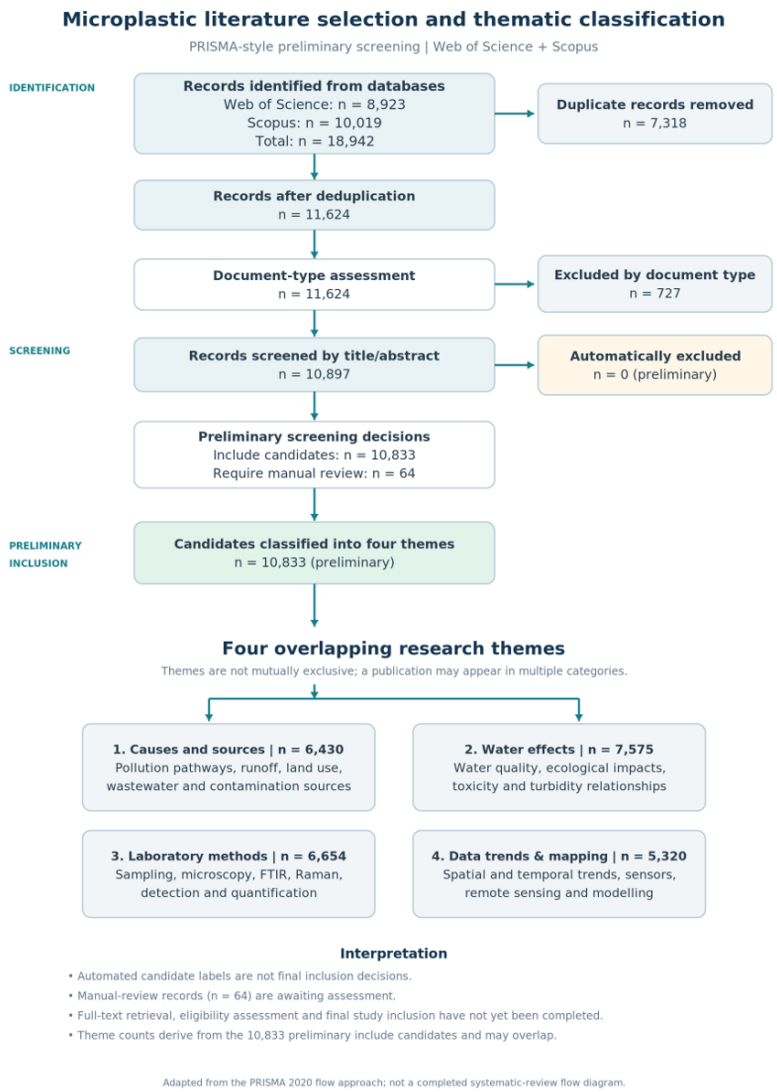


Figure 1: Preliminary PRISMA-style screening and thematic classification; final eligibility and full-text review remain outstanding.

Table 1: Top-level thematic classifications from 10,833 preliminary include candidates.

Theme	Papers tagged	Share of candidates
Causes and sources	6,430	59.4%
Water effects	7,575	69.9%
Laboratory methods	6,654	61.4%
Data trends and mapping	5,320	49.1%

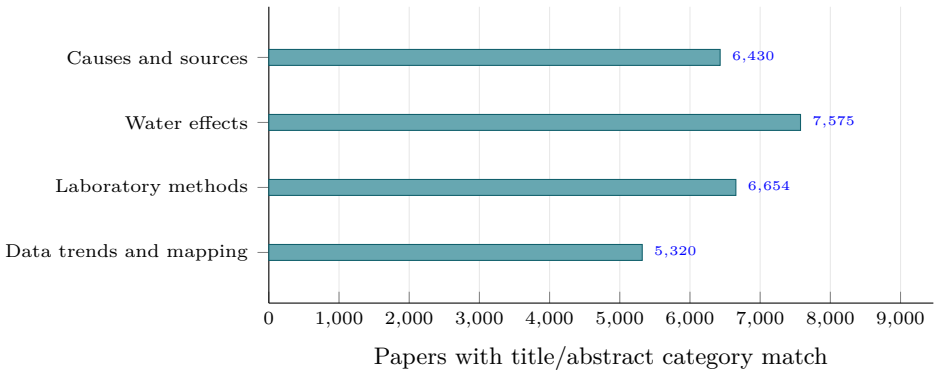


Figure 2: Four main research themes: automated, overlapping title/abstract subcategory mentions; not verified study findings.

Classification note. The Water Effects keywords reproduce the previously reported twelve-theme classification. The other three themes are newly subdivided using explicit keyword dictionaries supplied in the companion Python file. All four groups are multi-label, and both keyword omissions and false matches are possible. Percentages below use the number of candidates *within that parent theme*, not the overall 10,833 records.

2 Causes and sources of microplastics

The largest exploratory source tags concern plastic litter/degradation and wastewater. These labels indicate language in the papers, not source-apportionment results. To identify genuine inputs, quantitative source tracking, site characteristics, and reference sampling are needed.

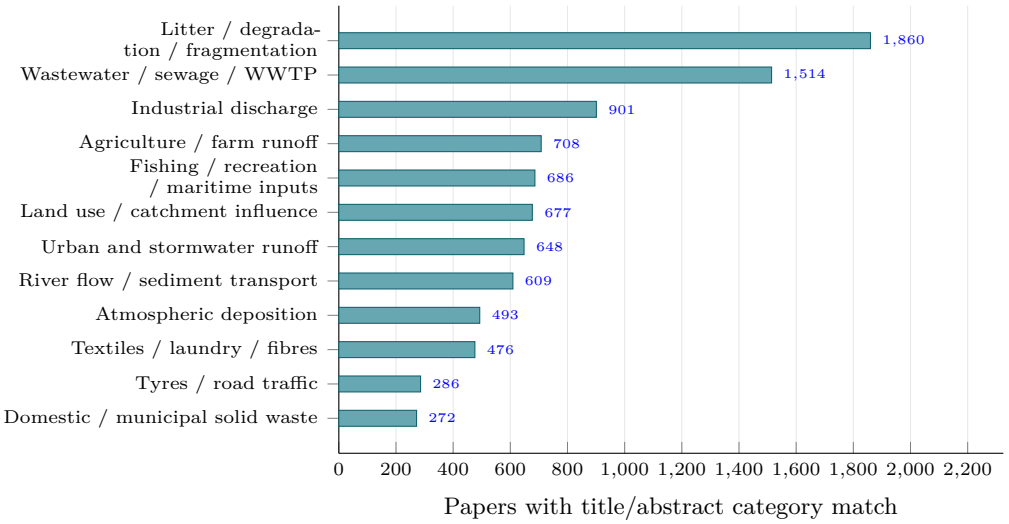


Figure 3: Causes and sources of microplastics: automated, overlapping title/abstract subcategory mentions; not verified study findings.

Table 2: Complete defined subcategory list within causes and sources of microplastics (n=6,430).

Subcategory	Tagged papers	Within-theme share
Litter / degradation / fragmentation	1,860	28.9%
Wastewater / sewage / WWTP	1,514	23.5%
Industrial discharge	901	14.0%
Agriculture / farm runoff	708	11.0%
Fishing / recreation / maritime inputs	686	10.7%
Land use / catchment influence	677	10.5%
Urban and stormwater runoff	648	10.1%
River flow / sediment transport	609	9.5%
Atmospheric deposition	493	7.7%
Textiles / laundry / fibres	476	7.4%
Tyres / road traffic	286	4.4%
Domestic / municipal solid waste	272	4.2%
No defined subcategory term matched	1,684	26.2%

Counts overlap; a paper can be in several rows. The unmatched count reflects this specific dictionary, not a screening exclusion.

Table 3: Illustrative study-level methodology and reported findings for causes and sources of microplastics. Findings are extracted/paraphrased from supplied abstracts and require full-text verification.

Study	Method or working principle	Abstract-reported observation / implication
Odora et al. [1]	Six Balu River surface-water stations near different industrial operations.	Mean 102.5 ± 12.83 items/L; site means differed. Observational site differences are not causal source attribution.
Fan et al. [2]	Ten Yangtze River sites; stereomicroscopy and micro-FTIR on water and sediment.	Surface-water abundance 46.7–204 particles/L; separate sediment data. Provides spatial/matrix context.
Di and Wang [3]	Reservoir surface-water and sediment survey with micro-Raman identification.	Water: 1,597–12,611 particles/m^3; different patterns were observed in sediments.

3 Effects on water and aquatic systems

The twelve established Water Effects labels span physicochemical measurements, biological endpoints, contaminant interactions, and exposure. They overlap and should not be interpreted as numbers of studies confirming adverse effects. For device design, the relevant subset is that reporting paired water-property and laboratory microplastic measurements.

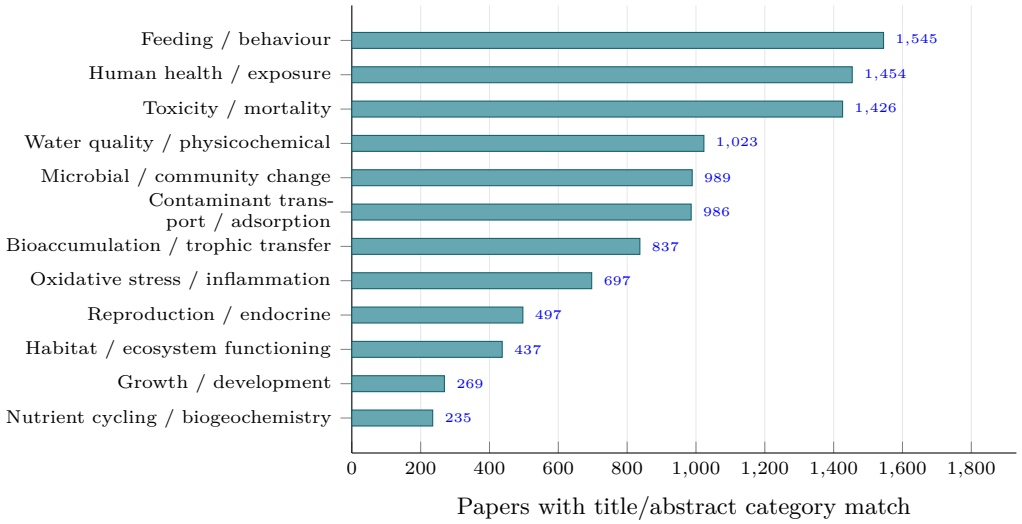


Figure 4: Effects on water and aquatic systems: automated, overlapping title/abstract subcategory mentions; not verified study findings.

Table 4: Complete defined subcategory list within effects on water and aquatic systems (n=7,575).

Subcategory	Tagged papers	Within-theme share
Feeding / behaviour	1,545	20.4%
Human health / exposure	1,454	19.2%
Toxicity / mortality	1,426	18.8%
Water quality / physicochemical	1,023	13.5%
Microbial / community change	989	13.1%
Contaminant transport / adsorption	986	13.0%
Bioaccumulation / trophic transfer	837	11.0%
Oxidative stress / inflammation	697	9.2%
Reproduction / endocrine	497	6.6%
Habitat / ecosystem functioning	437	5.8%
Growth / development	269	3.6%
Nutrient cycling / biogeochemistry	235	3.1%
No defined subcategory term matched	2,346	31.0%

Counts overlap; a paper can be in several rows. The unmatched count reflects this specific dictionary, not a screening exclusion.

Table 5: Illustrative study-level methodology and reported findings for effects on water and aquatic systems. Findings are extracted/paraphrased from supplied abstracts and require full-text verification.

Study	Method or working principle	Abstract-reported observation / implication
Liu and Li [4]	Statistical analysis of microplastics, nitrogen and electrical conductivity in Chinese surface waters.	Electrical conductivity was reported as a mediator (path coefficient 0.521); *not* a validated MP concentration sensor.
Sau et al. [5]	Sampling of 24 urban ponds/lakes with water-quality correlations and SVM classification.	Reported negative associations with turbidity/DO and positive associations with pH and TDS in that dataset; transfer to rivers is untested.
Rojas-Luna et al. [6]	Six lake stations across wet and dry seasons; filtration after digestion and density separation.	Abstract reports mean MP abundance 1.90 particles/L in rainy versus 0.25 particles/L in dry season.

4 Laboratory and measurement methods

The method groups describe techniques mentioned in the relevant titles and abstracts. A match does not establish that the technique was actually used in a study. Laboratory reference counting with polymer confirmation and contamination controls is needed to calibrate any proposed field proxy.

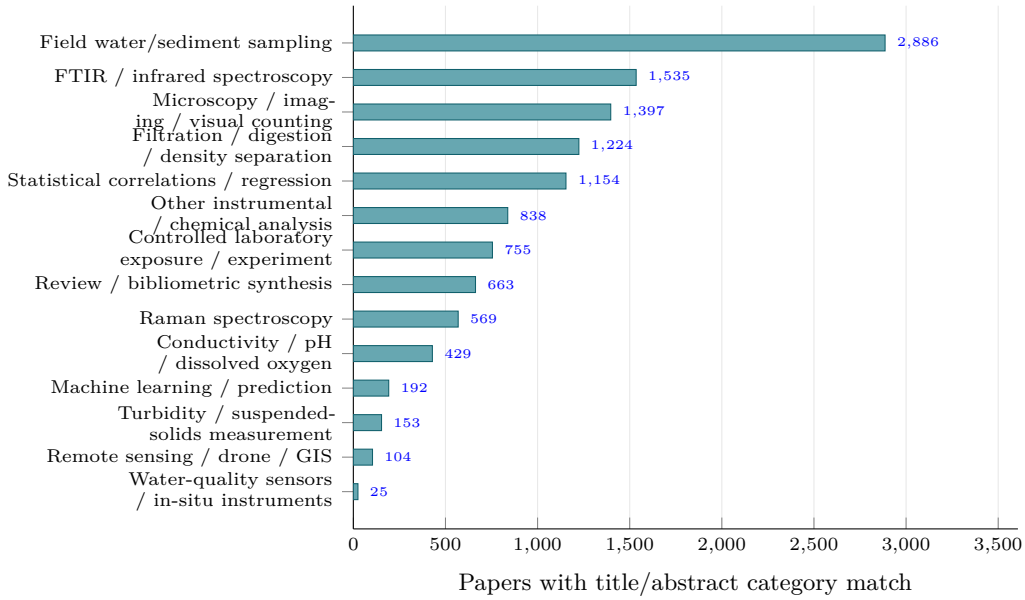


Figure 5: Laboratory and measurement methods: automated, overlapping title/abstract subcategory mentions; not verified study findings.

Table 6: Complete defined subcategory list within laboratory and measurement methods (n=6,654).

Subcategory	Tagged papers	Within-theme share
Field water/sediment sampling	2,886	43.4%
FTIR / infrared spectroscopy	1,535	23.1%
Microscopy / imaging / visual counting	1,397	21.0%
Filtration / digestion / density separation	1,224	18.4%
Statistical correlations / regression	1,154	17.3%
Other instrumental / chemical analysis	838	12.6%
Controlled laboratory exposure / experiment	755	11.3%
Review / bibliometric synthesis	663	10.0%
Raman spectroscopy	569	8.6%
Conductivity / pH / dissolved oxygen	429	6.4%
Machine learning / prediction	192	2.9%
Turbidity / suspended-solids measurement	153	2.3%
Remote sensing / drone / GIS	104	1.6%
Water-quality sensors / in-situ instruments	25	0.4%
No defined subcategory term matched	990	14.9%

Counts overlap; a paper can be in several rows. The unmatched count reflects this specific dictionary, not a screening exclusion.

Table 7: Illustrative study-level methodology and reported findings for laboratory and measurement methods. Findings are extracted/paraphrased from supplied abstracts and require full-text verification.

Study	Method or working principle	Abstract-reported observation / implication
Ermolin [7]	Review of water/sediment sampling, size/density separation and digestion methods.	Explains why sample preparation and recovery controls must be selected for the matrix; no single universal approach.
Rojas-Luna et al. [6]	72 water samples; H_2O_2 digestion, NaCl density separation, and filtration.	Observed abundance ranged from zero to 3.83 particles/L; laboratory extraction method affects the reportable fraction.
Fan et al. [2]	Digital stereomicroscope for particle counting and micro-FTIR for polymer identification.	Illustrates the distinction between suspected particle enumeration and polymer confirmation.
Narwal and Katyal [8]	Stereomicroscopy, electron microscopy and ATR-FTIR in surface waters.	Abstract reports 16–28 particles/L and polyethylene/polypropylene/polystyrene in analysed samples.

5 Data trends, monitoring and spatial mapping

This exploratory categorisation describes spatial, temporal, monitoring, modelling and imaging vocabulary. Studies predicting water attributes from images must be distinguished from studies estimating laboratory-confirmed microplastic concentrations with independent validation.

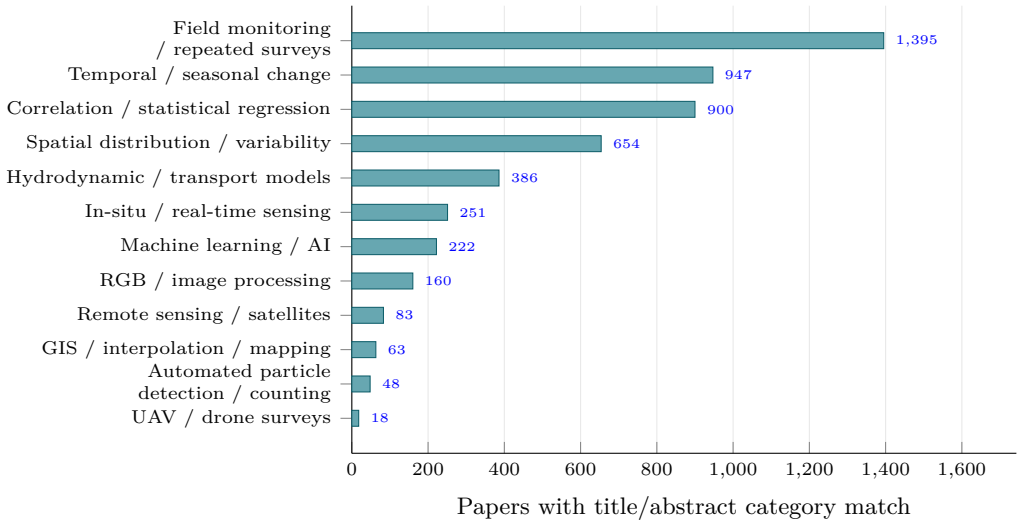


Figure 6: Data trends, monitoring and spatial mapping: automated, overlapping title/abstract subcategory mentions; not verified study findings.

Table 8: Complete defined subcategory list within data trends, monitoring and spatial mapping (n=5,320).

Subcategory	Tagged papers	Within-theme share
Field monitoring / repeated surveys	1,395	26.2%
Temporal / seasonal change	947	17.8%
Correlation / statistical regression	900	16.9%
Spatial distribution / variability	654	12.3%
Hydrodynamic / transport models	386	7.3%
In-situ / real-time sensing	251	4.7%
Machine learning / AI	222	4.2%
RGB / image processing	160	3.0%
Remote sensing / satellites	83	1.6%
GIS / interpolation / mapping	63	1.2%
Automated particle detection / counting	48	0.9%
UAV / drone surveys	18	0.3%
No defined subcategory term matched	2,030	38.2%

Counts overlap; a paper can be in several rows. The unmatched count reflects this specific dictionary, not a screening exclusion.

Table 9: Illustrative study-level methodology and reported findings for data trends, monitoring and spatial mapping. Findings are extracted/paraphrased from supplied abstracts and require full-text verification.

Study	Method or working principle	Abstract-reported observation / implication
Sau et al. [5]	Sentinel-2 remote sensing, GIS and support-vector-machine analysis.	Study reports morphology SVM accuracy of 89%; remote-sensing validation values are study-specific and require independent replication.
Chabot and Martein-son [9]	Review of aerial spectral, surface-roughness and indirect-proxy approaches.	Identifies approaches as largely experimental rather than established operational MP measurements.
Liu et al. [10]	Polarised scattering at 120° and machine-learning classification.	Experimental discrimination of polystyrene particles from algae/natural samples; not yet a calibrated river-wide concentration map.
Rojas-Luna et al. [6]	Repeated wet/dry season field sampling.	Seasonal difference illustrates why spatial maps need sampling date, rainfall and measurement uncertainty.

6 Compiled method–effect comparison and next decisions

Table 10: Connecting literature methods and effects to the proposed controlled-sample experiment.

Observed / studied relationship	Example literature evidence	What to validate in our chemistry laboratory
Spatial differences in particle abundance	Balu and Yangtze sampling [1], [2]	Replicate reference concentration by site and time; retain sample volume and GPS position.
MPs and nitrogen / conductivity	Surface-water statistical analysis [4]	Ionic strength and turbidity must vary independently of MP dose; assess whether adding MP improves prediction.
Wet/dry season variation	Repeated seasonal lake sampling [6]	Test model transfer across time and rainfall conditions.
Particle recovery and chemical identification	Sample-preparation review and stereomicroscope–FTIR surveys [2], [7], [8]	Quantify blank contamination, spike recovery, repeatability and polymer-confirmation rate.
Optical scattering and particle discrimination	Polarised-scattering experiments [10]	Challenge the sensor with algae, mineral sediments, differently sized polymers and mixed matrices.
Spatial prediction and aerial imagery	GIS/SVM study and aerial review [5], [9]	Train with georeferenced paired sensor–reference data; validate on completely withheld sites and days.

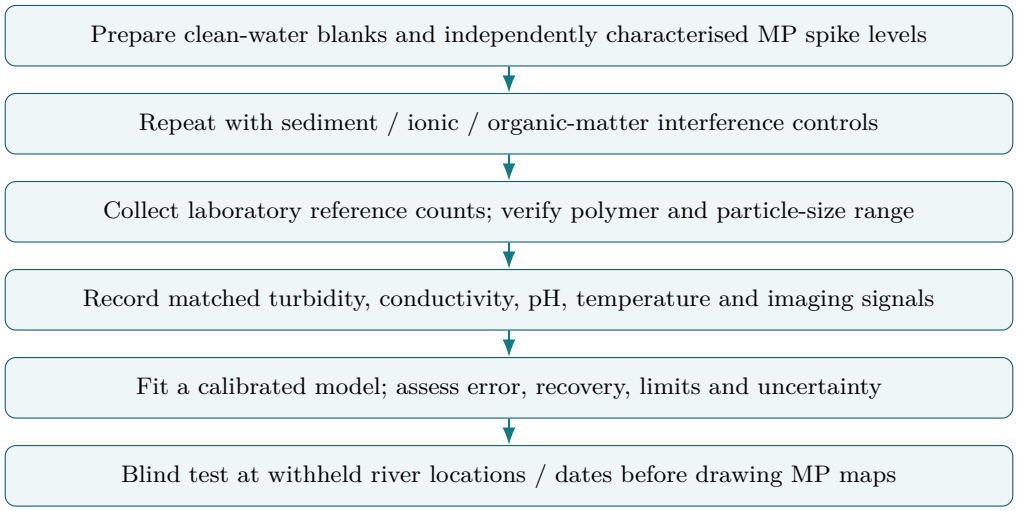


Figure 7: Proposed laboratory-to-field validation chain. This is a project plan, not a tested result.

7 Interpretation and limitations

The four top-level counts originate from preliminary screening, not final inclusion; 64 additional records await manual review. Subcategory labels are automated mentions in title/abstract text. Their counts may include reviews, descriptions of prior methods or hypotheses, and do not determine effectiveness, causality or evidence quality. The study examples illustrate different working principles but do not represent a quantitative synthesis of all candidate papers. Before final thesis conclusions, complete full-text screening, confirm reported effects and citations, and pre-specify the calibration and external-validation protocol with the chemistry laboratory.

References

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